

06 · RAP Research

Data Verification & How Commitments Vary

Indigenomics RAP Data Portal — Client Handoff Package

This document explains **two things we learned while building the platform** that shape how the data can and cannot be used:

1. **How we verify data sources** — the discipline that stops the AI from inventing facts, and how every figure on the dashboard traces back to a real, public document.
2. **How Reconciliation Action Plans (RAPs) actually vary** — in the words they use and, more importantly, in the *quality* of their commitments: many carry no due date and no measurable target, which has direct consequences for what a tracking platform can honestly report.

It closes with **how the platform is designed to cope with both**, and an honest account of where the limits are.

Companion documents: 01 Project Audit, 02 Deploy Runbook, 03 Engine Comparison, 04 Monitoring & Security, 05 Design Decisions & User Journeys. Unfamiliar terms (Bedrock, Lambda, BN, ...) are defined in the glossary in the package README.

1 What this document is based on (read this first)

Everything below is grounded in one of three kinds of evidence, and it matters which is which:

- **The data model and extraction rules** — how the software is *built* to handle RAPs. This is strong, first-hand evidence: the design was shaped, deliberately, around messy real documents.
- **Real spot-checks** — a small set of real, human-verified material: the **Bank of Canada RAP** (our one hand-verified “gold” set), a set of **hand-curated real figures** drawn from primary public sources (`real-fixtures.ts`), and the **live n=8 engine comparison** (see doc 03).
- **Demo data** — the ~100 organizations you see populating the running dashboard are **seeded illustrative data**, not extracted from real RAPs. They exist to demonstrate the product.

We are careful throughout to say “*the model and our real spot-checks confirm this,*” not “*we measured this across a large real corpus.*” We have **not** run the extraction pipeline across a large body of real RAPs — that requires the client’s own AWS account (see doc 02). The observations here are nonetheless well-grounded: where we make a claim about real RAPs, we point to a real example.

2 Part A · How we verify data sources

The platform holds two very different kinds of data, and “verification” means something different for each. Keeping them separate is the first honesty rule.

2.1 Two data regimes

- **The curated public-disclosure dataset** (behind the RAP Index dashboard). Each record is drawn from an **organization’s own public disclosure** — its reconciliation/ESG page or a first-party news release — and stores the source link. This is deliberately **not** sensitive Indigenous community data.
- **The AI extraction pipeline** (uploading a RAP PDF and reading commitments out of it). This is where the risk of AI “hallucination” lives, and where most of the verification machinery is aimed.

2.2 The core rule: the AI must *quote*, or the value is thrown away

The single most important design decision is this: **the AI’s job is to locate and quote, never to know or compute**. Every field the model extracts is wrapped in a structure that requires a **verbatim quote** and a **page number** from the source document. If the model cannot produce the exact words it read a value from, **the value is discarded** (set to null) rather than kept.

In plain terms: the system would rather report “*we didn’t find a due date*” than **invent** one. A confident, made-up citation is treated as worse than an honest gap.

2.3 A deterministic check proves the quote is really in the document

A quote alone isn’t enough — a model can fabricate a quote too. So after extraction, a piece of **ordinary, non-AI code** checks that the quoted text **actually appears in the document the model was shown**. It matches on words (so harmless whitespace or OCR punctuation differences don’t cause false alarms) but still catches fabricated or “welded-together” quotes.

This check earned its place. During development it caught a real, dangerous bug: on two-column pages, an earlier extraction produced **21 of 32 quotes that were fabricated** — plausible citations stitched from fragments, with off-by-one page numbers. The verbatim-quote gate is what now stops that class of error. On the validated run it produced **zero false positives across ~180 quotes**.

2.4 A human confirms before anything becomes official

Nothing an AI extracts reaches the live dashboard automatically. Every extraction lands in a **Pending Review** state; a human reviewer must confirm it (field by field) before it becomes a canonical record. And any number that matters — a target value, a due date, a percentage — is **parsed and calculated in code, never by the language model**. The AI finds the words; the software does the arithmetic.

2.5 Sources are ranked honestly: confirmed > research > self-reported

Not all data deserves equal trust, and the platform never pretends otherwise. Each record carries a **basis**:

- **Confirmed** — independently corroborated (e.g. procurement dollars attested by the supplier on the other side of the transaction). Only these can raise a commitment to “confirmed.”
- **Research** — curated by our team from an organization’s public disclosure. This is the default for the seed dataset.
- **Self-reported** — entered by a company about itself. These are **opt-in** and **do not count toward the headline figures** unless the organization’s identity is registry-verified and it chooses to be shown.

A self-report can **never** be silently promoted to “confirmed.” Identity itself is anchored to the organization’s **CRA Business Number** (with a checksum pre-filter and a registry lookup), so records attach to a real legal entity rather than a name that might be spelled three different ways.

2.6 The engine choice was measured, not guessed

Which AI reading engine to trust was settled by a **live, billed comparison across 8 documents** (full detail in doc 03). The recommendation — **Textract-LAYOUT** — won because it produced **real page numbers that were read from the document** (not inferred), solid quote grounding, and was the only engine to process all 8 documents without crashing. A cheaper text-layer engine is the residency-friendly fallback.

2.7 We proved that “let a second AI check the first” does not work here

A tempting shortcut is to have a second AI judge whether the first AI’s findings are correct. **We tested this and it failed.** Two independent judge models agreed with each other essentially only by chance (statistical agreement $\kappa \approx 0$) — because both simply assented to almost everything. On real, unlabeled RAP data, AI cross-checking gives **no reliable signal**. When humans adjudicated the handful of genuine disagreements, the split fell exactly on the *vague-and-inferred* vs *specific-and-grounded* line — which is precisely why a **human**, not an automated AI check, is the right reviewer. This is why the human-in-the-loop gate above is not bureaucratic caution; it is the only instrument that actually works.

2.8 The curated dataset is link-audited

For the public-disclosure dataset, **every source URL was HTTP-checked** — 102 of 102 unique links resolve to a live first-party page, with no fabricated URLs and no status ever inflated beyond what the source supports. Figures in that dataset are **illustrative snapshots** taken from the cited sources; the standing instruction is to confirm against the source before quoting a number as exact.

2.9 Honest limitations of the verification story

A client should know these plainly:

- **AI inference leaves Canada.** Data can rest in Canada, but Amazon’s Bedrock models are not hosted in Canada, so the reading step routes to a US/global region regardless of engine. (See doc 03; a Canadian-hosted model such as TELUS’s would close this gap.)
- **Our one human “gold” reference is a single document** (Bank of Canada). Broader accuracy is measured *relative to the combined output of all engines* — a defect every engine misses is invisible.

- **The production default engine (BDA) infers page numbers** rather than reading them, and is brittle on unusual PDFs. A back-fill step recovers verbatim quotes for it, but not trustworthy pages. Textract-LAYOUT is the recommendation precisely to avoid this.
- **Scanned / image-only PDFs are untested** — the corpus was all “born-digital,” and the text-layer engine has no OCR.
- **The business-registry (ISED) integration is coded but not yet activated** against the live API; today it runs as a stub.

3 Part B · How RAPs actually vary

Reconciliation Action Plans are **not** a standardized filing. They vary enormously — first in vocabulary, and more consequentially in the *substance* of what they commit to. This is the single biggest challenge for any platform that tries to track them, and it is worth the client understanding it directly.

3.1 They don’t even agree on what the document is called

Across real organizations, the same kind of artifact goes by different names:

- **Bank of Canada** — “Reconciliation Action Plan”
- **TELUS** — “Indigenous Reconciliation & Connectivity Report”
- **Suncor** — “Report on Sustainability — Indigenous Relations”
- **TC Energy** — “Indigenous Relations & Reconciliation”
- **Government of Canada** — “Mandatory Minimum 5% Indigenous Procurement Target”

The internal structure varies just as much: Bank of Canada organizes commitments into “**pathways**” (“People pathway,” “Learning pathway”); OPG uses a **5-pillar** structure; the Australian *Reflect RAP* format (Populous) mandates a rigid “**Action / Deliverable / Timeline / Responsibility**” table with maturity tiers (*reflect / innovate / stretch / elevate*) that have **no Canadian equivalent at all**.

How the platform copes: every commitment stores **both** the document’s own wording *and* a normalized value mapped onto a single canonical vocabulary (18 sectors, a fixed set of commitment types and themes). Nothing is lost — the original words are always kept beside the standardized one, so the dashboard can compare across organizations without erasing how each one actually spoke.

Be precise about how this works: the mapping from idiosyncratic wording to the canonical vocabulary is done by the **AI at extraction time and confirmed by the human reviewer** — it is not a fixed dictionary of synonyms. So terminology unification, like everything else, ultimately rests on the human-in-the-loop review gate.

3.2 The commitments themselves vary in structure

The extraction data model is built, deliberately, to **absorb** this variation rather than force every RAP into one shape:

- **Only two things are effectively required** of a commitment: an **action** and a **deliverable**.
- **Everything else is optional** and frequently absent: the **owner**, the **timeline / due date**, and any **quantitative target**.
- Sector-specific fields are filled in only when relevant, and a genuinely novel field surfaces in an “extras” bucket rather than being force-fit or invented — **absence is treated as meaningful information**, not an error.

3.3 Many commitments have no due date

This is real and common. A great many RAP commitments are phrased as an **ongoing cadence** (“Annual,” “Ongoing,” “Every three years”) or carry **no timeline at all**. Real examples from our verified material:

- **TELUS** — “*Maintain a public Indigenous reconciliation action plan*” (no date)
- **Agnico Eagle** — “*Publish a Reconciliation Action Plan*” (no date)
- The **entire Bank of Canada** commitment set is stored with no dates, because the document states none.

These are governance- or announcement-style statements. Read strictly, a commitment with **no date and no measure is not trackable** — there is no moment at which it can be said to be met or missed. From an accountability standpoint, an undated “maintain / publish / continue” pledge does little to demonstrate progress, however sincere the intent behind it. We note this as a **data-quality and accountability observation**, grounded in the documents themselves — not an accusation of motive.

(Design note: the platform stores the verbatim timeline wording next to the parsed date precisely so that a real cadence like “Annual” isn’t silently destroyed into “no timeline.” An earlier version made that mistake.)

3.4 Many commitments have no measurable target

Alongside missing dates, many commitments carry **no metric, KPI, or numeric target** — they are purely qualitative. Our verified examples (Bank of Canada, RBC, TELUS, Agnico) are dominated by qualitative action verbs: “*Develop a framework...*,” “*Engage in regular dialogue...*,” “*Seek guidance from...*,” “*Maintain...*” The Bank of Canada set is explicitly a set of “**qualitative pathways with no headline dollar targets.**”

A commitment with neither a target nor a date offers a tracking platform **nothing to track**. This is the crux of the accountability gap: the plan can be published and celebrated without ever exposing a checkpoint against which follow-through could be judged.

3.5 An honest caveat about these two patterns

The platform **records** whether a date or target is present (an empty value beside the preserved verbatim text), but it does **not yet flag or score** undated / unmeasurable commitments as a quality problem. In other words: the data to surface “how many of this organization’s commitments are actually trackable?” exists, but there is currently **no feature that presents it that way**. That is an honest gap — and, as Part C notes, a natural next step.

4 Part C · How the platform is designed to address this

Pulling Parts A and B together, here is how the product responds to messy sources and uneven commitments today, and where it can go next.

Already built:

- **Normalization that never erases the original.** Canonical vocabulary for cross-organization comparison, with each document’s own words retained alongside (B.1).
- **Grounding + human review** that absorbs structural variation safely: nullable-by-contract fields mean a RAP that omits owners, dates, or targets is handled honestly rather than hallucinated into completeness (A.2–A.4, B.2).
- **A progress and risk engine.** Commitments carry a status (on-track / delayed / met / missed / stalled) and an append-only history; the platform computes **overdue** and **at-risk** flags and produces plain-language digests (“N commitments are past their target year without confirmation”).
- **A confirmation-integrity view.** The dashboard measures the gap between what is *self-reported* and what is *independently confirmed* — the very gap the platform exists to surface — and never inflates a self-report into a confirmation.

Natural next step (not yet built):

- **A “trackability” signal.** Because the platform already records whether each commitment has a date and a measurable target, it could surface, per organization, *how many of its commitments are actually trackable* versus undated/unmeasurable — turning the observation in B.3–B.4 into a first-class, at-a-glance accountability indicator. This would be a modest, high-value addition built entirely on data the system already captures.

5 Part D · Evidence base & references

Real, verified material (safe to rely on):

- **Bank of Canada gold set** — `scripts/fixtures/gold-commitments-bankofcanada.json` (one human-verified RAP; the source of the “qualitative, undated” examples).
- **Hand-curated real figures** — `src/lib/rap/real-fixtures.ts` (real public figures drawn from primary sources by the team; real-world-true, but hand-curated, *not* produced by running the extraction pipeline).
- **Live engine comparison** — `docs/rap-engine-comparison.md` (billed n=8 run; the $\kappa \approx 0$ finding and the Textract-LAYOUT recommendation).
- **Extraction findings** — `docs/rap-extraction-findings.md` (the fabricated-quote bug, the verbatim-quote gate, over-extraction fixes).
- **Curated-dataset verification** — `DATA_VERIFICATION.md` (the 102/102 URL audit and the status-integrity rules).

- **Real RAP corpus (11 documents / ~250 pages)** — inventory in `Week 7/rap_samples/README.md`. These are readable source PDFs; extracting them at scale requires the client’s AWS account (doc 02), as the local demo returns canned data.

Where the mechanisms live in code (for a technical reader):

- The locate-and-quote contract and the canonical/optional field model — `src/lib/rap/types.ts`.
- The extraction rules (including “normalize themes” and “forward-looking only”) — `src/lib/rap/extraction-schema.ts`.
- The verbatim-quote gate — `src/lib/rap/validate.ts`.
- Source ranking (confirmed > research > self-reported) — `src/lib/index-evidence/resolver.ts`, `status-map.ts`.
- The overdue / at-risk risk engine — `src/lib/rap/insights.ts`.

Honesty summary for the client: the *design* evidence is strong and first-hand; the *real-RAP* evidence is small but genuine (one gold set + curated real figures + a live 8-document run). The breadth of variation visible in the running dashboard is mostly **seeded demo data**. Every claim in Parts A and B about real RAPs is backed by a real example above; none of it should be read as “measured across a large real corpus.”

Point-in-time handoff document. The living technical sources are in the repository at the paths cited above; the engine comparison (doc 03) is a live n=8 billed run, and the $\kappa \approx 0$ result reproduces a published “ecological boundary” finding — LLM cross-checking gives no decorrelated verification signal on this kind of real, unlabeled data, which is why human review is retained by design.